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Unit – 3B → Global Environmental Issues

Introduction

- There are significant changes in the global ecosystem after the World War II.
- The man-made forces disrupting the global ecosystem can be,
 - Population explosion:
 - demand and supply - by creating huge demands for shelter, food and various natural and manmade resources,
 - Industrialization and Urbanization:
 - resulted in more consumption of natural resources and more degradation of environment due to pollution.
- In the processes of developing, we are causing proportionate damage to the environment.
- This damage has led to problem at global scale, which namely includes,
 - Global warming/Climate change/Greenhouse effect
 - Depletion of ozone layer
 - Acid rain

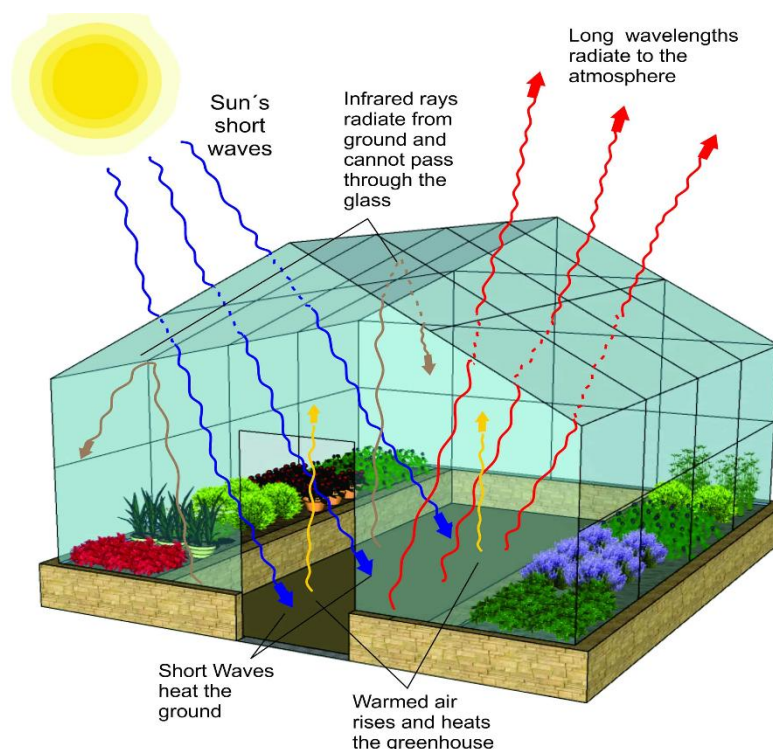
Global Warming

Introduction:

- Global warming simply means the overall increase in the average temperature of the whole globe (earth).
- Global warming can be understood, correlating the same with a phenomenon called greenhouse effect.

Greenhouse effect:

- Green house is a house made of glass to raise delicate plants and in cold region.
- Higher growth of plants inside the house is maintained by controlling high temperature, which is due to
 - Glass walls
 - High CO₂
 - Humidity
- These Glass walls of house allow the short-wave radiations from the sun but prevent the long wave infrared radiations coming out after reflection by earth inside the house.
- This causes an increase in temperature inside the house compared to outside.



Unit 3 – Environmental Pollution

- Some of the atmospheric gases and air pollutants have the property similar to the glasses.
- The atmospheric gases which are permeable to short wave radiation, but are strong absorber of infrared radiation (reflected from earth surface) is called greenhouse gases, which includes;
 - Carbon Dioxide (CO₂)
 - Nitrous Oxide (N₂O)
 - Methane (CH₄)
 - Chloro Fluoro Carbon (CFC)
- Among these CO₂ is mainly responsible for greenhouse effect, other gases are more potent than CO₂, but as their concentration is less, the effects of CO₂ is more compared to these gases.
- Global warming potential (GWP) is the ratio of the warming caused by substance compared to the warming caused by a similar mass of CO₂. GWP of CO₂ is 1 (obviously).
- **Global warming means increase in any temperature of the earth to greenhouse gases or overall avg. temperature of earth.**

Greenhouse gases:

- Carbon Dioxide (CO₂)
 - It contributes about 55-60% of the global warming caused by greenhouse gases produced by human activity.
 - It is produced by burning of fuels (67%), and other form of land clearing, burning, and deforestation (33%).
 - The level of CO₂ has increased from 280 ppm to 370 ppm in 2000.
- CFC
 - It contributes about 14-20% of global warming, from greenhouse gases.
 - CFCs are highly stable and nonflammable compounds of carbon and halogens and were used mostly as refrigerants, insulators and fire extinguishers.
 - It generally traps 1500 to 7000 time more heat molecule than CO₂.
 - The level of chlorofluorocarbons is around 0.00225 ppm, increasing at the rate of 0.5% annually.

- Methane
 - It contributes to 18-20% of global warming from greenhouse gases.
 - Its GWP is 25.
 - Methane gas is formed by the anaerobic decomposition of organic wastewater and sludge by anaerobic bacteria called methanogens.
 - It is produced by dumped garbage, wetlands, biomass burning.
 - The level of CH₄ has increased from 700 ppb to 1750 ppb in last the five decades.
- Nitrous Oxide
 - It is responsible for 6% of global warming, from green house gases.
 - Its GWP is 230.
 - Main source of N₂O is agriculture, biomass burning and industrial processes.
 - Atmospheric concentration of N₂O is 0.3 ppm, increasing at the rate of 0.2% annually.

Effects and Control of Global Warming

Effects of global warming:

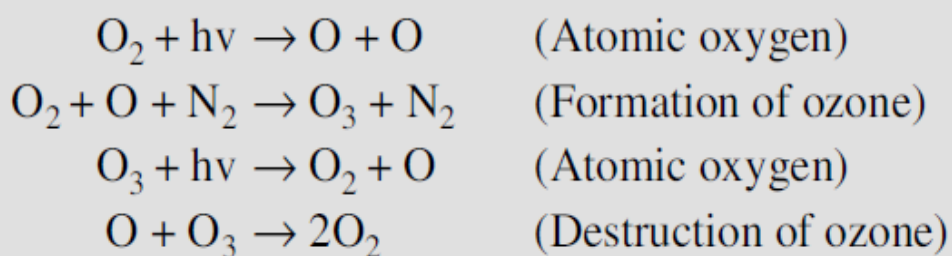
- Global average temperature of earth may increase by 1.4°C to 5.8°C by the year 2100 from 1990.
- In the 20th century temperature has increased about 0.6°C
- This increase in temperature causes
 - Melting of ice caps and glaciers
 - Rise in sea water was around 1 to 2 mm per year and if the rate of rise continues at least in the same way then in the next 100 years rise may be more than 20 cm.
 - Effects the rainfall pattern and is responsible for extreme drought or flood conditions.
 - Reduce crop production due to plant disease and pest growth.

Control of global warming:

- Reduction in the use of fossil fuel for energy generation and promoting other sources like solar energy, wind energy and nuclear energy.
- Minimizing the use of nitrogenous chemical fertilizers for agriculture.
- By reducing deforestation in planting more and more trees as they trap air pollutants and utilize CO₂.
- Trapping and using methane as fuel.
- Using energy resources judiciously.
- Stabilizing population growth.

Ozone layer Depletion

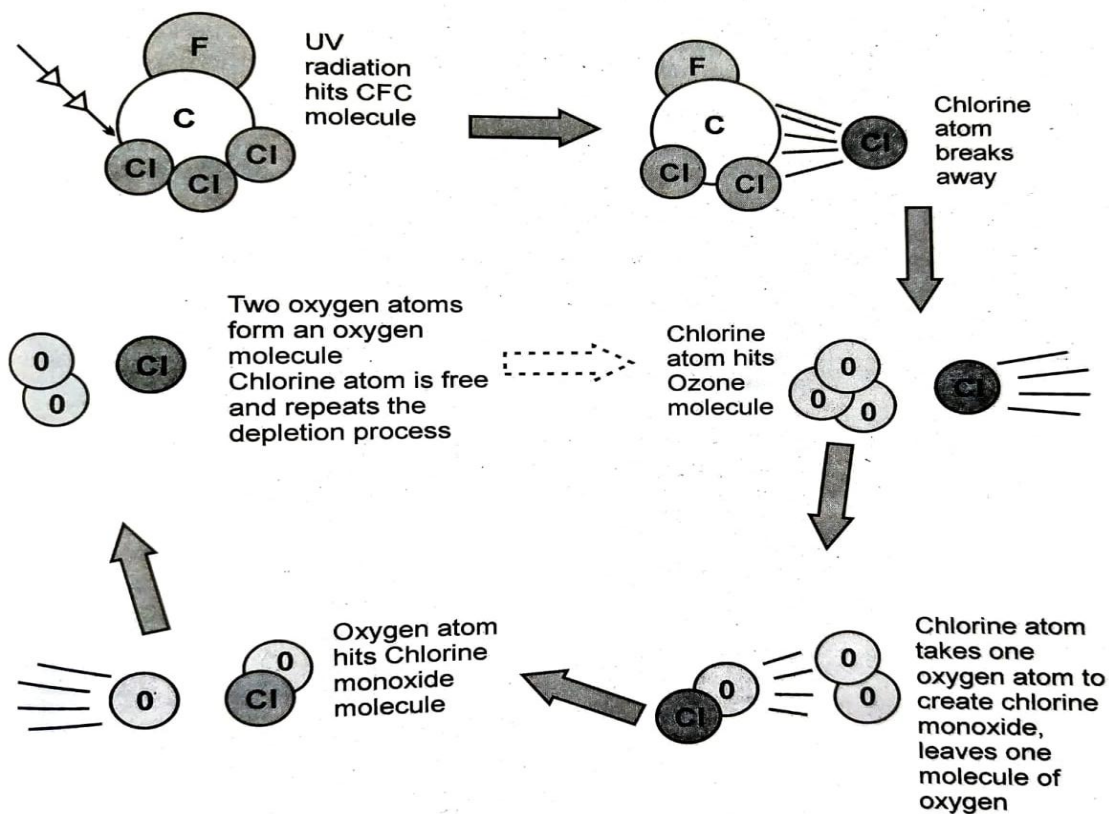
- A layer of O₃ is present in the stratosphere at the height of 20-25 km above the ground level.
- This layer filters out harmful ultraviolet radiation from the sunlight, thus protects various life forms on earth.
- It acts like natural sunscreen for earth.
- Around 90% of the total atmospheric ozone is generally present in the stratosphere as ozone layer while the remaining 10% is found in the troposphere.
- The formation of ozone in the stratosphere is cyclic as it is continuously formed as well as destroyed as given in the figure.'



- Ultraviolet rays decompose oxygen into atomic oxygen by photolytic decomposition.
- The atomic oxygen formed rapidly reacts with molecular oxygen to form ozone.
- Ozone thus formed is distributed in the region and absorbs UV radiation and over time gets converted into molecular oxygen.
- In the time of 1970 warning of depletion was declared, the first incidence of ozone layer depletion was in 1985, over Antarctica.

Destruction of Ozone:

- Main chemical responsible for ozone destruction is CFCs & CF₃Br (Chlorofluorobromines).
- High concentration of NO₂, if present in stratosphere (by the emissions of Jet planes) may also destroy ozone as given below.



Chlorofluorocarbon:

→ CFC is a combination of carbon, hydrogen, fluorine, and chlorine.

→ CFC – 11 and CFC- 12 are most commonly used CFCs.

→ Uses of CFC are;

- Used as a coolant in refrigerators and air conditioners.
- Propellant.
- Cleaning solvent.
- Stuffing of mattresses.

→ ODP means ozone depletion potential which is ratio of the impact on ozone caused by a chemical compared to the impact of a similar mass of CFC-11.

Measurement of Ozone concentration:

→ Unit of measurement of ozone in atmosphere is Dobson unit (DU).

→ One Dobson unit (DU) is defined as 0.01 mm thick ozone layer at STP (i.e., 0°C and 1 atm).

Unit 3 – Environmental Pollution

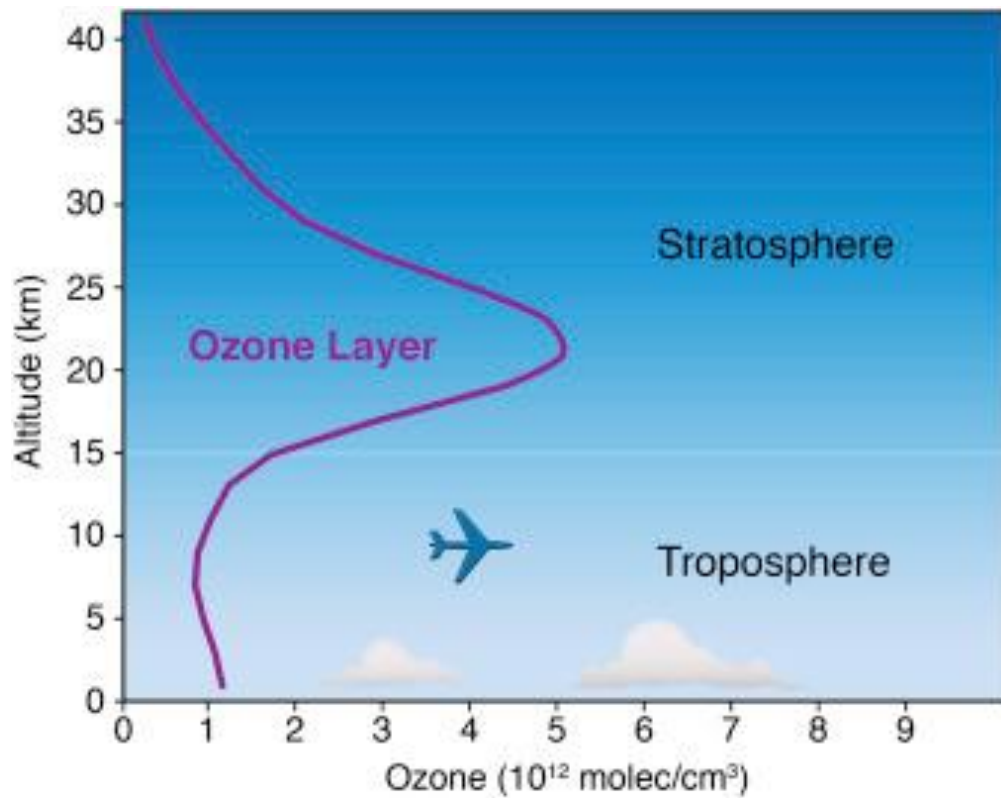
→ Under normal conditions, earth atmosphere contains about 300 DU i.e., 3 mm thick slab of ozone layer which is reduced to 100 DU, i.e., 1 mm thick ozone slab, at ozone holes.

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Global Average Ozone: 300 DU=3 mm



Ozone Hole Average: 100 DU=1 mm



Effects & Control of Ozone depletion

Effects of ozone depletion:

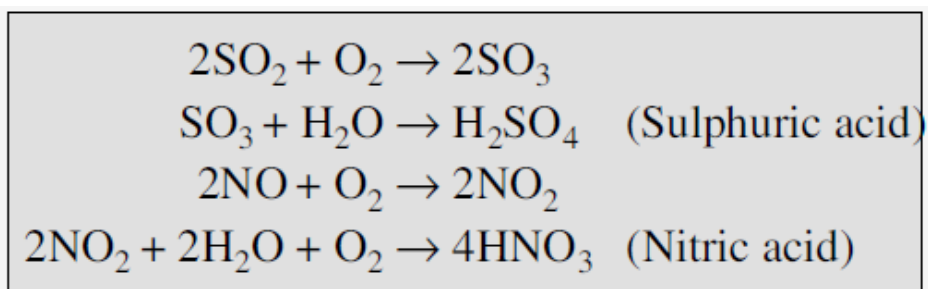
- Ozone depletion cause more UV-B radiations to reach on to the earth which affects DNA and causes skin cancer as well as affects the lens and cornea of the eye.
- Ozone depletion may cause low productivity and reduction in photosynthesis.
- It may reduce the growth of phytoplankton, zooplanktons and small fishes by disturbing the aquatic ecosystem.
- More UV radiations due to ozone depletion can also cause degradation of paints, plastics and polymers, etc.

Control of ozone depletion:

- Use of HCFC instead of CFC can reduce Ozone depletion.
- Due to the presence of hydrogen bond, HCFC is less stable and reactive compared to CFC.
- Ozone depletion potential of HCFC is about 0.01–0.5

Acid Rain

- When the pH of the rain water becomes less than 5.6, then it is called acid rain.
- This reduction of pH is basically due to the dissolution of oxides of nitrogen and sulphur present in the atmosphere which forms nitric and sulphuric acids.



- It can be caused by oxides of carbon, sulphur, and nitrogen.
- Sulphuric acid (60-70%) and nitric acid (30-40%) are the major acid found in rain water.

Causes of Acid rain:

- SO₂ and NO_x are mainly responsible for forming of acid rain, main sources of such pollutants are;
 - Fossil fuel burning
 - Smelting of sulfide ore
 - Automobile exhaust
 - Industrial plant using sulphuric acid and nitric acid

Environmental effects of acid rain

Effects of acid rain on material:

- Acid rain can damage the structural materials like marble and lime stone. It may corrode and decolorize the walls of monuments and thus deteriorate their beauty.
- Many metals get oxidized. Iron corrodes with the presence of acid rain to form rust.
- It deteriorates paints.

Effects of acid rain on living organism:

- Aquatic life especially fishes are badly affected by lake acidification.
- Acid rain mobilizes heavy metals such as calcium and mercury in soil and rocks and sediments, which are then leached out by rain and enter the surface water.
- Acidified lakes have high level of cadmium, lead, aluminum, manganese, zinc, copper and nickel.
- All these can kill living organism if present in sufficient quantities.

Effects of acid rain on fertility of soil:

- Due to high solubility of acidic rain water, plants nutrients like nitrogen, phosphorous, and potassium gets leached away, which reduces fertility of soil.
- It damages tree growth.
- It makes trees more susceptible to stress like cold temperature, drought, etc.
- Many insects and fungi are more tolerant to acidic conditions and hence they can attack the susceptible trees and cause disease.

Control of acid rain:

- Emission of SO₂ and NO_x from industries and power plants should be reduced by using pollution control devices.
- Addition of lime in, lakes and soils should be done to correct the adverse effect of acid rain.